

Exploring the socioeconomic drivers of deforestation in Bangladesh: The case of Teknaf Wildlife Sanctuary and its surrounding community

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ABSTRACT

Bangladesh is experiencing deforestation within its protected forest areas (PAs). One reason for its inability to stem deforestation is the lack of a proper understanding of the deforestation process and the drivers responsible for it. This study aims to elucidate the socioeconomic factors influencing deforestation within a PA on the coast of Bangladesh known as Teknaf Wildlife Sanctuary (TWS). In the quest to understand the drivers of deforestation, the extent of community involvement in the deforestation process was explored. The local community was grouped based on their involvement in various deforestation drivers. These groups were then compared to determine the socioeconomic factors influencing the likelihood of their involvement in deforestation. The community survey covered 10% of households in the community surrounding TWS and involved interviewing 5769 household heads. This study found that 2150 households (37% of all surveyed households) were engaged in fuelwood collection, 623 (11%) were encroaching on the TWS, and 506 (9%) were cultivating betel leaves. Therefore, 45% of households were engaged in some form of deforestation driver. Among the socioeconomic factors, new households with small properties had a higher likelihood of being involved in deforestation drivers. Moreover, comparatively poor households had a significantly higher probability of being engaged in deforestation. Regarding occupations, farmers and laborers had a higher likelihood of being associated with deforestation. This study sheds light on the complex interplay of socioeconomic factors influencing the deforestation process occurring within a specific PA in Bangladesh under existing forest policies. This understanding can be useful for policymakers and forest managers in developing effective forest policies and managing forests more sustainably.

1. Introduction

Forests are critically important for the existence of human civilization. Nearly 2 billion people live within a 10-km radius of forests, while approximately 1 billion live within a 1-km radius of forests (Newton et al., 2020). Approximately one-third of the world's 7.8 billion people rely on forests and their resources for livelihood and other necessities (FAO and UNEP, 2020). Forests are a source of food, fuel, fiber, and income for millions of people, and forest-based industries provide employment and revenue for many others (Angelsen et al., 2014). Besides contributing directly to the livelihoods of millions of people, forests regulate climate, sequester carbon, and serve as a habitat for the majority of terrestrial biodiversity (Hajjar et al., 2020; Houghton et al., 2015; Newton et al., 2016). Despite offering such benefits, natural forests are decreasing. During the 2015–2020 period, 10 million hectares of

forests were destroyed every year (FAO and UNEP, 2020). Humans are also responsible for forest loss by converting forest lands into other forms of land (McGowan, 2016; Venter et al., 2016; Wang et al., 2019). Therefore, people's relationships with forests constitute a significant area of interest among researchers, policymakers, and governments. In the context of declining biodiversity, people-forest relationships have become increasingly significant and a prerequisite to promoting sustainable development.

Protected areas (PAs) have been a major spearheading measure aimed at protecting forests and their biodiversity. In 2020, there were 258,608 PAs covering 20.28 million square kilometers of land, which is nearly 15% of the earth's land surface (UNEP-WCMC, 2020). Despite the recent increase in PA coverage, there has been a growing body of evidence indicating widespread forest degradation, wildlife habitat loss, and even the collapse of species within PAs (Brodie and Waterhouse,

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2012; Geldmann et al., 2019; Watson et al., 2014). To stop deforestation and ensure the effectiveness of PAs, the first step involves understanding the causes of deforestation and the drivers responsible for them. Several meta-analyses have attempted to quantify the impact of common drivers of deforestation at a global scale (Curtis et al., 2018; Ferrer Velasco et al., 2020; Hosonuma et al., 2012). While developing and implementing national-level forest conservation policies, it is also necessary to develop a local-level understanding of the drivers of deforestation. Besides the identification and quantification of national- and local-level drivers of deforestation, recognizing the socioeconomic dimensions associated with the causal mechanism of the deforestation process is indispensable.

Bangladesh is a developing country in the tropical region. Overall, 12.8% of its land is covered by forests (Henry et al., 2021). Despite having one of the lowest amounts of forests per capita (0.009 ha/person), Bangladesh has lost 40% of its forests since 1930 and has seen a reduction of 106 square kilometers each year between 1990 and 2015 (Reddy et al., 2016; FAO, 2015). Most of the natural forests in Bangladesh are state-owned and managed by the Bangladesh Forest Department (BFD). To protect natural forests, BFD has adopted the “establishing of PAs” as its main strategy and marks boundaries around the remaining patches of natural forest. The average area of parks and sanctuaries that are considered PAs fall within the size range of 500 to 15,000 ha, with some of the few exceptions including large mangrove forests in the southern coastal areas (BFD, 2020). Currently, Bangladesh has 58 PAs, among which more than 30 have been established since 2000 (BFD, 2020). However, most of the PAs are experiencing continuous deforestation within their boundaries (Ahmed et al., 2020; Islam et al., 2019; Rahman and Islam, 2021a). Since most of the forests are owned by the state, the BFD is the key authority tasked with establishing and enforcing management and conservation policies aimed at protecting the remaining natural forests of Bangladesh.

Much like many other developing countries, the BFD had a recent paradigm shift in its forest policies, with the management of forests changing from centralized management to community management, where local communities are included in the forest management and decision-making processes (Nath et al., 2016; Speer, 2012). The BFD has since adopted new forest policies and strategies that focus on increasing the number of PAs and implementing community management approaches for governing new PAs. This new strategy defines forest management as co-management, where local communities and the BFD co-manage protected forests in a participatory manner. Since 1990, more than 150 forest-related policy developments and changes have occurred in Bangladesh (Rahman and Giessen, 2017). Of these changes, 17 were related to the formulation and implementation of major forest rules, with policies being enacted and adopted to facilitate the implementation of forest co-management (Rahman et al., 2016). However, this paradigm shift in forest management must be based on an in-depth understanding of the relationship between forests and local people. Studies have found that anthropogenic pressures from local communities are causing deforestation within and around PAs in Bangladesh (Islam et al., 2021; Tani and Rahman, 2017). Therefore, new policies and strategic frameworks should address this issue based on an accurate understanding of the existing drivers of deforestation.

This study focuses on the deforestation process of a PA known as the Teknaf Wildlife Sanctuary (TWS) in Bangladesh. Previous studies have identified several drivers linked to the local community that are responsible for deforestation within the TWS (Alam et al., 2015; Ullah and Tani, 2017; Tani and Rahman, 2017; Ullah et al., 2020a). However, there is a lack of in-depth understanding regarding how these drivers are influenced by the socioeconomic attributes of the local community. Since the government of Bangladesh intends to include more local communities in forest conservation and management efforts, the existing relationships between the local communities and adjacent PAs must be analyzed. As a result, this study pursues the following objectives: 1) to describe the socioeconomic profiles of the local communities engaged in

different drivers of deforestation; 2) to determine the attributional difference observed between the households directly engaged in deforestation and those that are not directly engaged; 3) to elucidate the socioeconomic determinants that impact the likelihood of households functioning as the drivers of deforestation. The findings of this research will contribute to a greater understanding of the deforestation process and the involvement of the local community in this process within Bangladesh. This can serve as an important step toward developing forest management policies in PAs and their surrounding communities in Bangladesh.

2. Theoretical concept

Deforestation can be defined as the removal of native vegetation. Deforestation is multidimensional, context-specific, and caused by several social processes. To elucidate the processes and causes of deforestation, it is important to understand the context of forest cover change. This section will discuss two theories: 1) forest transition: where the relationship between the state of society and forest cover change is conceptualized and 2) the causal mechanism of forest cover change with a focus on the deforestation process.

Forest transition (FT) is a well-established concept for understanding the nature of forest cover change at the national level. Notably, it is correlated with demographic and economic aspects. FT was first explained by Mather (1992) as forested land having turning points that ranged from shrinking to expanding, which are triggered by factors including population trends, industry, the market economy, and public interests. As FT evolved, it was adopted to consider more specific factors related to national economic development, with the transition being broadly divided into two phases: deforestation and reforestation (Barbier et al., 2010; Grainger, 1995; Lambin and Meyfroidt, 2010; Perz, 2007). According to the FT theory, a country loses forests (deforestation) at the initial stage of economic development, which eventually stops at a certain point and the loss begins to reverse (reforestation).

FT has subsequently been tested in several developing countries, and it was found that forest cover at the national level followed an inverse J-shaped curve over time (Kauppi et al., 2006; Mather, 2007; Rudel et al., 2005). Upon considering 100 developing countries, Hosonuma et al. (2012) suggested that there are four phases of FT: pre-transition (high forest cover and low deforestation), early transition (high deforestation), late transition (low forest cover with slowing deforestation), and post-transition (forest cover increase). According to Hosonuma et al. (2012), Bangladesh is in the late transition phase. A recent report showed that the rate of forest cover decrease has been reduced. When considering tree coverage from 2000 to 2014, the forest cover of Bangladesh increased by approximately 136,000 ha (Potapov et al., 2017; The World Bank, 2018). Although forest cover increased at the national level, the ongoing high deforestation rate in PAs and natural forests remains a serious concern for Bangladesh. To conserve the PAs of Bangladesh, it is important to understand the causal process of deforestation.

While the causal mechanism of deforestation is complex and region-specific, a substantial number of deforestation drivers have been reported worldwide. The actions or activities that cause deforestation are known as the drivers of deforestation. These drivers are interlinked and multilayered within a nested structure and collectively contribute to deforestation (Tani and Rahman, 2017). Since the 1980s, numerous attempts have been made to explain the causative pattern of deforestation (Curtis et al., 2018; Geist and Lambin, 2001; Hosonuma et al., 2012; Mainardi, 1998). Therefore, Geist and Lambin (2001) proposed a simplified structure of the drivers of deforestation, which is among the most widely accepted multi-causation theories used to describe the deforestation process (Fig. 1). This approach describes the deforestation process as a consequence of the interlinked execution of proximate and underlying drivers.

Proximate drivers are human activities that directly affect the

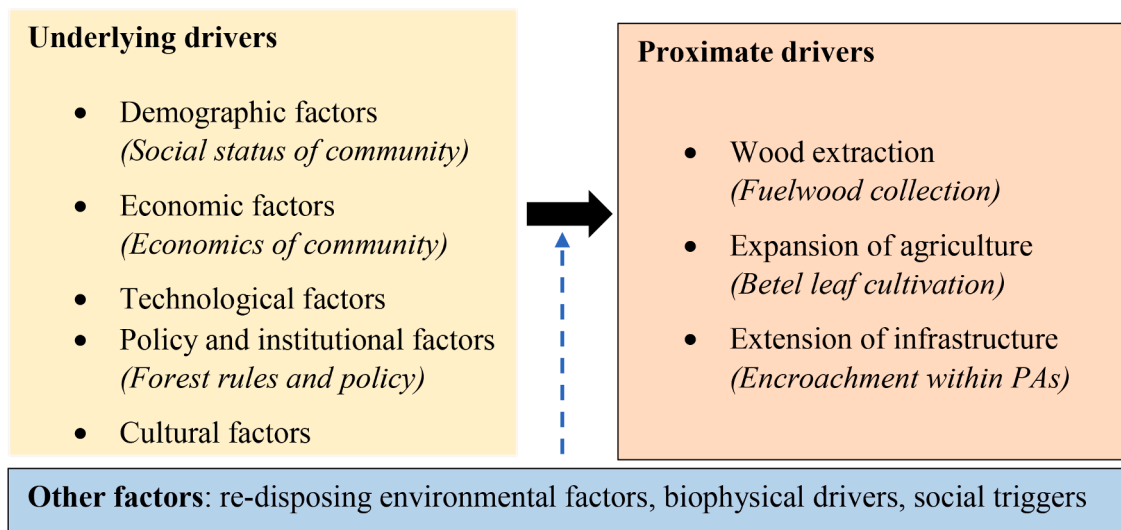


Fig. 1. A simplified version of the deforestation framework adopted from Geist and Lambin (2001); italicized text represents the issues discussed in this study.

deforestation process. These are categorized as agricultural expansion, wood extraction, and infrastructure extension. The underlying drivers are fundamental social processes that underpin the proximate causes, which comprise a complex of social, political, economic, technological, and cultural variables (Geist and Lambin, 2001). While proximate drivers operate at a local scale, the underlying drivers may operate at local, national, or even global scales. By using the theory of Geist and Lambin (2001), deforestation in the TWS can be described as a regional case. Previous studies have identified the proximate drivers of deforestation (e.g., betel leaf cultivation, fuelwood extraction, and encroached households within the protected forest), which resemble the proximate driver categories of agricultural expansion, wood extraction, and

infrastructure extension, respectively (Ullah and Tani, 2017; Alam et al., 2015; Asahiro et al., 2013; IPAC, 2009; Sakamoto and Tani, 2013; Tani, 2017a, ; Tani et al., 2013, 2011; Ullah et al., 2020a; Ullah and Tsuchiya, 2017). As per the deforestation theory, these proximate drivers are influenced by several underlying drivers. However, it is more difficult to describe the impact of underlying factors on deforestation than deforestation and its proximate causes.

This study attempts to elucidate the impact of the underlying drivers—including demographic factors, socio-political factors, and economic factors—on the proximate drivers to generate a better understanding of the deforestation process.

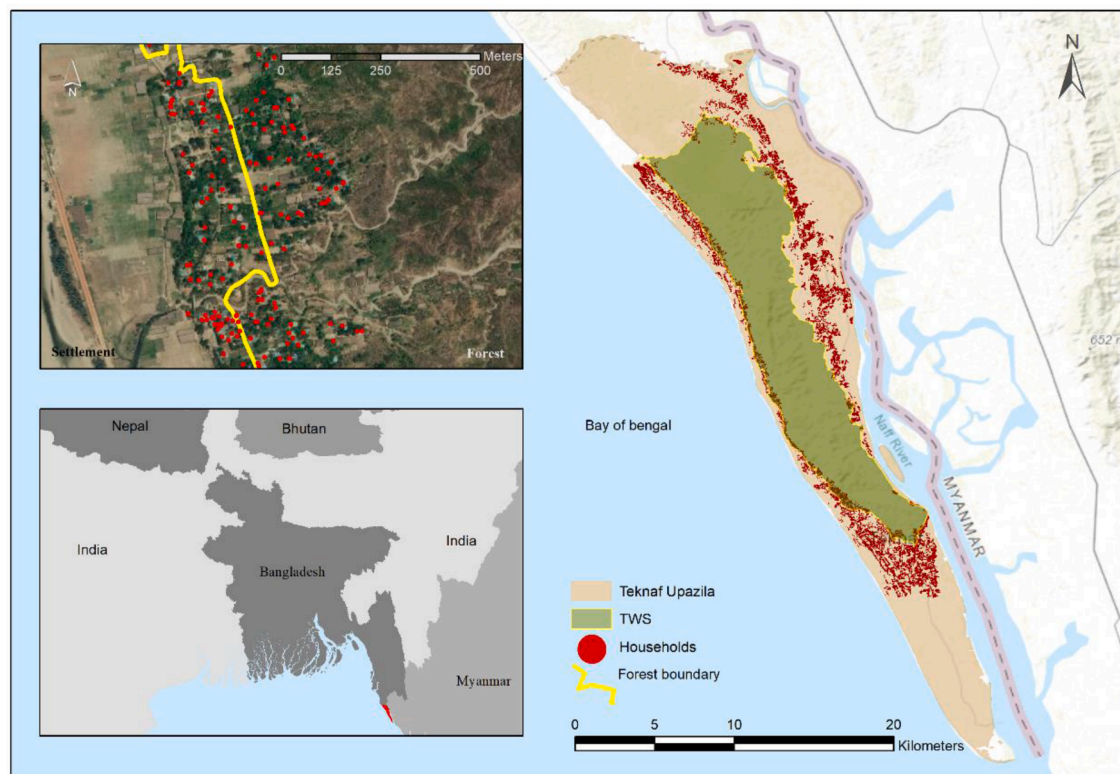


Fig. 2. Study area showing Teknaf Upazila, TWS, and the surrounding local community.

3. Methodology

3.1. Study area

This study focuses on a protected forest area known as TWS (see Fig. 2). TWS is located in Teknaf Upazila, which is in the southern district of Cox's Bazar. Upazila is an administrative unit (sub-district) under the district administration and comprises several grassroots-level administrative units known as unions. Teknaf Upazila is located at 20°51'56" N and 92°17'43" E, which is nearly 48 km south of Cox's Bazar district. Teknaf Upazila is surrounded by water bodies on three sides, namely the Bay of Bengal on the west and south, and the Naf River on the east. This region is also known as the Teknaf Peninsula. Administratively, Teknaf Upazila is comprised of six unions: Teknaf Sadar, Baharchhara, Whykhong, Nhila, Sabrang, and St. Martin—the latter being an island located a few kilometers from the shore (not considered in this study). Each of the unions comprises 11 to 48 villages. The total population of Teknaf Upazila is 366,979, 42% of which is living in poverty (HIES, 2016). The local community is highly dependent on forests and their resources to sustain daily living (Tani and Rahman, 2017). In Teknaf Upazila, forest land represents a total of 14,602 ha. Of this area, 11,615 ha is authorized as a protected area (i.e., TWS). The TWS stretches 28 km from north to south and 3 to 5 km from east to west. TWS is rich in biodiversity, and the forest is broadly classified as mixed tropical evergreen and semi-evergreen. TWS is famous for being one of the last few habitats for Asian elephants (*Elephas maximus*) (Alam et al., 2012). Besides the elephants, Indian civets are among the list of endangered species found in the forests of TWS. Notably, TWS is one of the oldest PAs in the country—holding the status of a reserved forest since 1907—and was formerly known as the Teknaf Game Reserve (TGR). Despite holding reserved forest status for a long period of time, TWS has continuously been losing forest area due to the anthropogenic pressure received from the local community. Its forests are currently in critical condition and on the verge of losing the capacity to host biodiversity, which has resulted in TWS becoming a PA and there being a need to obtain a better understanding of its situation to avoid further deforestation.

3.2. Data collection: study population, sampling, and survey

The population targeted for this empirical study was the entire population of Teknaf Upazila (except St. Martin Island). Existing records for the local community were obtained from the Bangladesh Bureau of Statistics (BBS). According to the BBS (2011), there are approximately 46,000 households spread over 153 villages in Teknaf Upazila. This study followed a quantitative approach and used face-to-face interviews to survey the sampled households. The data collection survey and data management were conducted during the 2016–2017 period. First, the population was listed by counting and listing all of the households within the targeted study area. This study included 57,404 households in Teknaf Upazila. A systematic sampling technique was used to select the households for the survey. The target was to survey 10% of the households in Teknaf Upazila. Thus, every 10th household in Teknaf Upazila was selected for the survey, and a total of 5769 households were surveyed. During the household survey, a pre-structured questionnaire was used for the face-to-face interviews held with household heads. This socioeconomic survey followed the methodology described by Angelsen et al. (2014) to measure the livelihoods and environmental dependence of people. Since a large number of households were surveyed across nearly 150 villages, local survey assistants (6 to 10) were recruited to conduct surveys after receiving proper training.

3.3. Data analysis

3.3.1. Variables and analysis

The household interviews collected data on household head (HH)

age, education, family size, and occupation. Education was determined by the number of formal schooling years, while family size was defined as the number of family members living together. Occupation was determined based on the occupations of the HHs, which served as their main occupation. While the members of many households engaged in several livelihood activities, most of them responded spontaneously about their main occupation choice. If the HHs could not respond, we asked them about the occupation they engaged in the most themselves or which occupation they thought was the most important in terms of their social identity or income generation. Data on household tenure, annual income, and land property were also collected. Tenure refers to the number of years a family has been living in their current house. Land refers to the total land accessible by a household in hectares. Annual income was determined based on the 1-year recall period of the total gross income of a household in Bangladeshi currency, which is known as Taka (BDT).

Both descriptive and inferential statistical methods were adopted to analyze quantitative data via the use of SPSS Statistics for Windows, version 16. Descriptive statistics such as mean values, categorization, frequency counts, and percentages were used to describe the socioeconomic characteristics of the local community. With regard to the categorization and analysis, households were grouped based on their association with the drivers of deforestation, such as betel leaf cultivation, fuelwood collection, and encroachment within the TWS. If the members of a household were cultivating betel leaves, they were categorized as a group of betel leaf cultivators. When the family members of any household collected fuelwood from the restricted TWS, they were categorized as a group of fuelwood collectors. With respect to the case of encroachment, a house was considered an encroached household if it was situated within the forest boundary. These household groups were not mutually exclusive. And household members could engage in more than one activity that functioned as a driver of deforestation. Another household group was formed by including households that engaged in at least one activity that functioned as a driver of deforestation, which was termed the “driver” group. Households that did not directly engage in any activity that functioned as a driver of deforestation were categorized as the “non-driver” group. To draw comparisons between the different groups, a one-way sample *t*-test was performed to determine the mean differences among the groups and their statistical significance. Additionally, the quantitative variables were categorized into various sets to elucidate the socioeconomic characteristics of the households.

3.3.2. Logit model specification

Furthermore, to assess the impact of the socioeconomic determinants on the probability of households engaging in deforestation activities (drivers), four binary logistic regression models were developed. The binary logistic regression model is a feasible statistical method used when the outcome of a dependent variable is binary in nature (Cramer, 2003; Hosmer et al., 2000). The model is based on the concept of cumulative logistic probability and is commonly used to determine the relationship between a binary response variable and explanatory variables, with the relationship being either discrete or continuous (Jitmun et al., 2020; Kabir et al., 2021). Here, the households' engagement in activities that functioned as drivers of deforestation was coded as a binary choice (yes = 1, no = 0) and used as a dependent variable. The logit models contained seven explanatory variables, including age, tenure, family size, education, land, annual income, and occupation. The explanatory variables were categorized into different groups, and these groups were considered discrete variables (Table 3). For example, based on annual income, the households were categorized into a poor group, a non-poor group, and an above poverty group. These groups were then treated as discrete explanatory variables against the dependent variables.

In the binary logit model, the logit is the natural logarithm (ln) of odds *Y*, while the odds are the ratios of the probabilities (π) of *Y* occurring to the probabilities ($1-\pi$) of *Y* not occurring. The logit model is

as follows:

$$\text{logit}(Y) = \ln\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki}$$

where β_0 is the intercept and $\beta_1, \beta_2, \dots, \beta_k$ are the coefficients of the independent variables $X_1, X_2, \dots, X_k$.

The dependent variables in the logit models were the households' likelihood of being associated with the drivers of deforestation. Four separate logit models were developed to predict the likelihood of households engaging in the activities of collecting fuelwood, encroaching within the forest boundary, cultivating betel leaves, or any other driver of deforestation. If the households were associated with the drivers of deforestation, they were coded as 1, while the remaining households were coded as 0. Three logit models were developed based on encroachment, fuelwood collection, and betel leaf cultivation. To interpret the models, the estimated coefficients' and odds ratios' significant parameters were used. Positive or negative coefficients indicated the direction of impact of the determinants. A unit change in the determinants resulted in a change in the odds ratios. Based on this change, the predictions of the binary logistic regression model could easily be interpreted. Odds ratios of > 1 indicated an increase in the likelihood of an event's occurrence, while odds ratios of < 1 indicated a decrease. An important condition of the binary logit model is the lack of multicollinearity among the variables (Cramer, 2003). Prior to the logit regression, multicollinearity was tested for the explanatory variables by checking the variance of the inflation factor (VIF).

4. Results

4.1. Socioeconomic profiling of households related to the drivers of deforestation

Table 1 presents the socioeconomic status of the local communities associated with the various drivers of deforestation. This study found that 2150 households (37% of the surveyed households) were engaging in fuelwood collection, 623 (11%) were encroaching within the TWS,

and 506 (9%) were cultivating betel leaves. As previously discussed, these activities constitute the major causes of deforestation in the Teknaf Peninsula, and the households associated with these drivers are directly responsible for the continuous forest loss observed. The socioeconomic status of the households related to the various drivers of deforestation will generate a better understanding of the nature and role of the local community in the deforestation process. Regarding the age of HHs, the average age was 44.47 years (12.52 SD) among betel leaf cultivating HHs. The average age of the HHs associated with fuelwood collection and encroachment was approximately 40 years (13 SD). Since the average age ranged from 40 to 44 years across all households, most of the HHs (76 to 82%) belonged to the middle-aged group. The average tenures of fuelwood collecting households, encroacher households, and betel leaf cultivating households were 14.11 (19.93 SD), 10.58 (9.69 SD), and 18.07 years (22.10 SD), respectively. Of the fuelwood collecting and encroacher households, the houses of 40% of fuelwood collecting households and 43% of encroacher households were new, meaning that they were established within the last 5 years. The betel leaf cultivating households had a higher tenure compared to the other households, with 45% of the households having a tenure of 5 to 15 years.

The family size of the households averaged around six members (2.20–2.73 SD). The highest average family size was that of the betel leaf cultivating households, with the average being 6.91 members per family. A total of 50% of the households had family sizes of more than six members. In the cases of the fuelwood collecting households and encroacher households, 39 and 37% of the respective households had family sizes of more than six members. Regarding the average number of schooling years, the fuelwood collecting households, encroacher households, and betel leaf cultivating households had averages of 1.61, 1.63, and 1.29 years, with standard deviations ranging from 2.73 to 2.91. Among the betel leaf farmers, 76% received no formal education. The educational backgrounds of the fuelwood collectors and encroachers were comparatively better, with 67% of the fuelwood collectors and 66% of the encroachers receiving no formal education.

With regard to land property, 89% of the fuelwood collecting households, 84% of the encroacher households, and 74% of the betel

Table 1
Socioeconomic profiles for different drivers of deforestation.

Variables	Category N=	Deforestation drivers Fuelwood collection 2150	Encroachment 623	Betel leaf cultivation 506
Age (years)	Mean $\pm$ SD	40.75 $\pm$ 13.01	39.51 $\pm$ 12.89	44.47 $\pm$ 12.52
	Young (0–29)	19	23	12
	Middle-aged (30–60)	76	72	82
	Old (> 60)	5	6	7
	Mean $\pm$ SD	14.11 $\pm$ 19.93	10.58 $\pm$ 9.69	18.07 $\pm$ 22.10
Tenure (years)	New (0–5)	40	43	21
	5 to 15	34	36	45
	16 to 30	19	18	24
	Old (> 30)	7	3	10
	Mean $\pm$ SD	6.14 $\pm$ 2.36	6.05 $\pm$ 2.20	6.91 $\pm$ 2.41
Family size (number of individuals)	Small (< 5)	25	26	13
	Medium (5 to 6)	36	37	37
	Large (> 6)	39	37	50
	Mean $\pm$ SD	1.61 $\pm$ 2.94	1.63 $\pm$ 2.94	1.29 $\pm$ 2.73
	None	67	66	76
Education (school years)	Up to primary	25	26	18
	High school	6	7	4
	University	2	2	1
	Mean $\pm$ SD	0.11 $\pm$ 0.23	0.15 $\pm$ 0.27	0.23 $\pm$ 0.40
	Landless (< 0.2 ha)	89	84	74
Land (ha)	Marginal (0.21–0.50 ha)	7	9	16
	Small (0.51–1.0 ha)	3	5	7
	Medium (1.1–2.0 ha)	1	2	3
	Large (> 2.0 ha)	0	0	1
	Mean $\pm$ SD	169 $\pm$ 254	163 $\pm$ 143	213 $\pm$ 195
Annual income (000 BDT)	Poor (below avg. expenditure)	68	69	52
	Non-poor (poor – 1.9 USD)	8	8	8
	Above poverty (above 1.9 USD)	24	23	40

leaf cultivating households possessed no land, with the average land property possessed being less than 0.20 ha. The betel leaf cultivating households possessed the most average land property (average: 0.23 ha). The average land property possessed by the fuelwood collecting households and encroacher households were 0.23 and 0.27 ha, respectively, with 11–16% of the respective households not being landless. When it came to the betel leaf cultivating households, 27% were above the landless category.

The average annual income was highest for the betel leaf cultivating households (213,000 BDT). The average annual income of fuelwood collecting households was 169,000 BDT, while that of the encroacher households was 263,000 BDT. Furthermore, the households were categorized based on income as either poor, non-poor, or above poverty households. The poor households' average monthly income was below rural Bangladesh's average monthly expenditure, which was 14,156 BDT per month and 169,872 BDT annually, considering the national average family size (HIES, 2018). Households with a per capita income of more than 1.90 USD per day (i.e., annual income of 199,167 BDT per household considering the average family size) were categorized as above poverty households. Households between the poverty and above poverty categories were considered non-poor. Approximately 68% of the fuelwood collecting households and encroacher households were poor, and only about 24% of them were above the 1.90 USD daily poverty line. Betel leaf farmers had slightly better parameters, with 54% of them being poor and 40% being above the poverty line.

4.2. Comparison of the drivers and non-drivers of deforestation

This section provides a comparison of the socioeconomic characteristics of the drivers and non-drivers of deforestation. Table 2 presents a comparison of the average values for the socioeconomic characteristics of drivers and non-drivers based on an independent sample *t*-test. HH average age was approximately 40 years, with no significant difference

in HH age between the drivers and non-drivers. The highest frequency was found among the middle-aged group, which was 76% for both the drivers and non-drivers. When considering tenure, a significant difference was found between the drivers and non-drivers. The average tenure of drivers was 14.02 years (19.51 SD). The non-drivers had a lower average tenure of 12.64 years (14.65 SD). The average tenure of the local community (considering all the households) was 13.26 (17 SD). Most of the houses were in the 5 to 15 years old category, constituting approximately 44 to 48% of the houses in the community (among drivers and non-drivers).

Regarding the new households (0–5 years of tenure), drivers had a slightly higher frequency (31%) than non-drivers (29%). There was no significant difference in the family sizes of households between the drivers and non-drivers; however, the drivers had a slightly larger family size (6.19 members) in comparison to non-drivers (6.11). Overall, the average family size of the local community was 6.14 members (2.42 SD). While the medium-sized family group had a frequency of approximately 36%, the large-sized family group had a frequency of approximately 38%.

The Teknaf region is not among the most progressive regions in the country. Nevertheless, 65% of all HHs with no formal schooling were the subjects of concern in this study. The HHs' overall average schooling years was 1.97 years (3.46 SD). Moreover, there was a significant difference between the drivers and non-drivers with regard to the number of schooling years. Driver HHs had fewer schooling years (1.65 years) in comparison to the non-driver HHs (2.22). Among all HHs, approximately 65% had no formal schooling, while only 23% had completed primary education equivalent to 5 years of formal schooling. Regarding land property, the local community households had an average land area of 0.11 ha. There was no significant difference between the drivers and non-drivers with regard to land property ownership. However, 88% of the households had less than 0.2 ha of land and were considered landless. Only 1% of the households had more than 2 ha of land property

Table 2
Comparison of the drivers of deforestation and non-drivers.

Variables	Category	N =	Deforestation drivers		
			All 5769	Drivers 2573	Non-drivers 3196
Age (years)	Mean $\pm$ SD		40.23 $\pm$ 12.52	40.88 $\pm$ 12.89	39.71 $\pm$ 12.20
	Young (0–29)	%	19	19	19
	Middle-aged (30–60)		76	76	76
	Old (> 60)		5	5	5
Tenure** (years)	Mean $\pm$ SD		13.26 $\pm$ 17.00	14.02 $\pm$ 19.51	12.64 $\pm$ 14.65
	New (0–5)	%	29	31	27
	5 to 15		46	44	48
	16 to 30		19	18	20
	Old (> 30)		6	7	5
Family size (number)	Mean $\pm$ SD		6.14 $\pm$ 2.42	6.19 $\pm$ 2.37	6.11 $\pm$ 2.45
	Small (< 5)	%	26	24	27
	Medium (5 to 6)		36	37	36
	Large (> 6)		38	39	37
Education** (school years)	Mean $\pm$ SD		1.97 $\pm$ 3.46	1.65 $\pm$ 3.02	2.22 $\pm$ 3.76
	None	%	65	68	63
	Up to primary		23	23	23
	High school		10	8	11
	University		2	1	3
Land (ha)	Mean $\pm$ SD		0.11 $\pm$ 0.30	0.12 $\pm$ 0.26	0.11 $\pm$ 0.33
	Landless (< 0.2 ha)	%	88	88	89
	Marginal (0.21–0.50 ha)		6	7	5
	Small (0.51–1.0 ha)		3	3	3
	Medium (1.1–2.0 ha)		2	1	2
	Large (> 2.0 ha)		1	1	1
	Mean $\pm$ SD		195 $\pm$ 227	175 $\pm$ 244	210 $\pm$ 211
Annual Income** (000 BDT)	Poor (below avg. expenditure)	%	61	66	58
	Non-poor (poor – 1.9 USD)		9	8	9
	Above poverty (above 1.9 USD)		30	26	33

** Significant mean difference between the drivers and non-drivers. Independent sample T-test results are presented as (variable: t-value, p-value) age: –3.516, > 0.05; tenure: –2.985, < 0.01; family size: 1.243, > 0.05; education: 6.436, < 0.01; land: –1.084, > 0.05; annual income: 5.725, < 0.01.

Table 3
Model to predict the likelihood of deforestation drivers.

	All drivers Est.	Odds ratio	Fuelwood collection Est.	Odds ratio	Encroachment within the forests Est.	Odds ratio	Betel leaf cultivation Est.	Odds ratio
Age (years)								
Young (0–29)	0.488	1.431 ³	0.105	1.111	0.298	1.347	1.340	3.819 ³
Middle (30–60)	0.358	1.629 ²	0.017	1.017	–0.053	0.948	1.067	2.907 ³
Old (> 60) ref.								
Tenure (years)								
New (0–5)	–0.144	0.866	–0.110	0.896	0.896	2.450 ³	–1.026	0.358 ³
5 to 15	–0.500	0.606 ³	–0.508	0.602 ³	0.674	1.961 ²	–0.543	0.581 ²
16 to 30	–0.501	0.606 ³	–0.371	0.690 ²	0.611	1.843 ¹	–0.528	0.590 ¹
Old (> 30) ref.								
Family size								
Small (< 5)	–0.165	0.848 ¹	–0.191	0.826	–0.077	0.926	–0.246	0.782
Medium (5 to 6)	–0.055	0.947	–0.141	0.869	–0.006	0.994	0.078	1.081
Large (> 6) ref.								
Education								
No	–0.001	1.118	0.043	1.044	0.166	1.181	0.098	1.103
Up to primary	0.173	0.999	0.329	1.389 ²	0.318	1.374	–0.122	0.885
Higher education ref.								
Land								
Landless (< 0.2 ha)	0.649	1.913 ²	0.513	1.669 ²	–0.315	0.730	0.351	1.420
Marginal (0.21–0.50 ha)	0.959	2.609 ³	0.653	1.922 ²	0.297	1.345	0.823	2.278 ¹
Small (0.51–1.0 ha)	0.464	1.590	0.288	1.334	0.221	1.247	0.285	1.330
> Small ref.								
Annual income								
Poor	0.200	1.222 ²	0.316	1.371 ³	0.465	1.592 ³	–0.470	0.625 ³
Non-poor	0.059	1.061	0.112	1.119	0.305	1.357	–0.331	0.718
Above poverty ref.								
Occupation								
Abroad	–0.175	0.840	–0.147	0.863	0.124	1.132	0.043	1.044
Business person	–0.214	0.807 ¹	–0.265	0.767 ²	–0.007	0.993	0.907	2.478 ²
Fisherman	0.176	1.192	0.326	1.385 ²	–0.251	0.778	0.241	1.272
Farmer	1.891	6.624 ³	1.109	3.031 ³	0.711	2.035 ³	4.271	71.69 ³
Laborer	0.741	2.099 ³	0.800	2.226 ³	0.379	1.460 ²	0.780	2.181 ¹
Service ref.								
–2 Log likelihood	7230.94		3826.37		7178.26		2107.56	
Prediction accuracy	65.1%		89.2%		65.7%		91.7%	

¹ $p < 0.05$.

² $p < 0.01$.

³ $p < 0.001$.

and were considered large landowners. Annual income is one of the most significant parameters used for describing the socioeconomic status of a community. The average annual income of the local community was 195,000 BDT. However, there were significant differences in annual income between the drivers and non-drivers. Non-drivers had a higher annual income (210,000 BDT) than the drivers (175,000 BDT). Overall, 61% of all households were poor. Moreover, a total of 66% of the drivers and 58% of the non-drivers were poor. Only 26% of the drivers were above poverty. This was slightly higher in the case of non-drivers, with 33% of non-driver households having a daily income of more than 1.90 USD per head.

4.3. Logistic model to predict the socioeconomic factors impacting the drivers of deforestation

The results of the logistic model are presented in Table 3. All of the four logit models considered showed an adequate fit of the model for data in the Hosmer and Lemeshow goodness-to-fit test, with the P value being greater than 0.05 and the prediction accuracy ranging from 65.1 to 91.7%.

The prediction probability is described based on the odds ratio and the estimated coefficient. Here, while the odds ratio determines the extent of the prediction, the coefficient determines the direction of the prediction. In these logit models, age, tenure, family size, education, land property, annual income, and occupation were considered determinants of the likelihood of households engaging in any of the drivers of deforestation. Here, all these drivers indicate the possibility of households engaging in any activities that function as deforestation

drivers. Fuelwood, encroachment, and betel leaf cultivation indicate a household's possibility of engaging in activities such as collecting fuelwood, encroaching within the protected forest area, and cultivating betel leaves, respectively.

4.3.1. Logit model for all drivers of deforestation

With regard to the case of the logit model for all of the drivers, while predicting the likelihood of the households engaging in any of the activities functioning as the drivers of deforestation, the coefficients of age, tenure, family size, land annual income, and occupation were found to be statistically significant. Considering the variables of age, households with young and middle-aged heads had 43 (odds ratio 1.431) and 63% (odds ratio: 1.629) greater likelihoods of contributing to forest conversion than households with old heads. In both cases, the coefficients were positive and significant. Concerning the variable of tenure, households with tenure periods of 5–15 years and 16–30 years had significant and negative coefficients. In both cases, the households had a 39% lower likelihood (odds ratio: 0.606) of associating themselves with any of the drivers of deforestation than the households with a tenure period of more than 30 years. Regarding family size, small families had a 15% (odds ratio 0.848) lower likelihood of engaging in any of the drivers of deforestation when compared to large families (more than six members). Overall, small families had a negative and significant coefficient. Education had no significant coefficient while pre-dating the likelihood of engaging in forest conversion activities. Meanwhile, with regard to land property, landless and marginal households had significant and positive coefficients. Landless households were 1.9 times more likely (odds ratio: 1.913) and marginal households were 2.6 times more

(odds ratio: 2.601) likely to engage in any of the activities functioning as the drivers of deforestation compared to the households with more than 1.0 ha of land property. Based on annual income, the households were grouped as poor, non-poor, and above poverty households. Poor households had significant positive coefficients, with a 22% greater likelihood (odds ratio: 1.222) of engaging in forest destruction activities when compared to the above poverty households.

Occupation choice had a significant impact on predicting the likelihood of a household engaging in forest conversion activities. Concerning occupation, while business persons had significant negative coefficients, farmers and laborers had significant positive coefficients. In comparison to those providing services, business persons were 21% less likely (odds ratio: 0.807), laborers were 2.1 times more likely (odds ratio: 2.099), and farmers were 6.6 times more likely (odds ratio: 6.624) to engage in activities that function as the drivers of deforestation.

4.3.2. Logit model for fuelwood collectors

The logit model used to predict a household's likelihood of collecting fuelwood found households to have statistically significant coefficients for tenure, education, land property, annual income, and occupation. With regard to tenure, households with tenure periods of 5–15 and 16–30 years had significant and negative coefficients. These households were nearly 40 (odds ratio: 0.602) and 31% (odds ratio: 0.690) less likely to collect fuelwood in comparison to households with a tenure period of more than 30 years. Regarding education, HHs with primary education were 39% (odds ratio: 1.389) more likely to collect fuelwood in comparison to HHs with education higher than the primary level. Concerning land property, the logit model suggested that the households holding less land property had a higher probability of collecting fuelwood, with significant positive coefficients. Landless and marginal households were 67 (odds ratio: 1.669) and 92% (odds ratio) more likely to collect fuelwood than those with more than 1 ha of land property. Regarding annual income, households had significant coefficients. Poor households had a 37% (odds ratio: 1.371) greater chance of collecting fuelwood in comparison to households living above the poverty line. Concerning occupation, except the ones abroad, others had significant coefficients while predicting the likelihood of households collecting firewood. Business households had a 23% (odds ratio: 0.767) lower chance in comparison to service households. Fishing, labor, and farming households were 1.4 (odds ratio: 1.385), 2.2 (odds ratio: 2.226), and 3.0 (odds ratio: 3.031) times more likely to collect fuelwood than service households.

4.3.3. Logit model for encroachers

The logit model for encroachment predicted the factors influencing the likelihood of households engaging in encroachment. The model found these households to have statistically significant coefficients for the variables of tenure, annual income, and occupation. In the logit model, the likelihood of households with new houses encroaching on protected forests was higher than that of households with old houses. Households with a tenure of 0 to 5 years were 2.5 times (odds ratio: 2.450)—while those with a tenure of 5 to 15 years were 2.0 times (odds ratio: 1.961)—more likely to encroach on the forests than the households with old houses. Poor households were 1.6 times more likely to encroach than households above the poverty line. Farming and labor households had significant and positive coefficients and were 2.0 (odds ratio: 2.035) times and 1.5 times (odds ratio: 1.460) more likely to encroach on the forest boundary.

4.3.4. Logit model for betel leaf cultivators

While developing the logit model to predict the determinants of the likelihood of households engaging in betel leaf cultivation, the household coefficients for age, tenure, land property, and occupation were statistically significant. With regard to HHs, young and middle-aged heads were 3.8 (odds ratio: 3.819) and 2.9 (odds ratio: 2.907) times more likely to cultivate betel leaves in comparison to old household

heads. The households had statistically significant and negative coefficients for tenure. New households and households with a tenure of 5 to 15 years were 74 (odds ratio: 0.358) and 42% (odds ratio: 0.581) less likely to cultivate betel leaf, respectively, in comparison to households whose tenure period was more than 30 years. Households with marginal lands (0.21–0.50 ha) were 2.3 times more likely to cultivate betel leaves in comparison to large households. Upon predicting the likelihood of cultivating betel leaves based on annual income, poor households had a 37% lower (odds ratio: 0.625) likelihood of doing so in comparison to households living above the poverty line. Regarding households' main occupations, farming households were nearly 72 times (odds ratio: 71.69) more likely to cultivate betel leaves in comparison to service households. Moreover, business households were 2.5 times (odds ratio: 2.478)—and labor households were 2.2 times (odds ratio: 2.181)—more likely to cultivate betel leaves in comparison to service households.

5. Discussion

Efforts to understand the relationship between deforestation and various socioeconomic processes gained popularity among academics during the 1980s. However, these efforts can be traced back to the 1870s when [von Thunen \(1875\)](#) published his classic work on the economic geography of forests. von Thunen described how the socioeconomic context influences deforestation and reforestation processes. Since then, with the continuous urge to understand deforestation, famous deforestation theories and frameworks ([Kuznets, 1955](#); [Ostrom, 1990](#); [Geist and Lambin, 2001](#); [Rudel et al., 2005](#)) have been proposed to describe the influence of social and economic elements on the deforestation process. Based on the understanding of these theories and frameworks, this study focused on elucidating the influence of socioeconomic aspects on the deforestation process in Bangladesh at a local scale.

5.1. Forest drivers and non-drivers

Our findings underscore the impact of the various socioeconomic aspects of households on the deforestation process in the Teknaf Peninsula at the local level. The data presented in [Table 1](#) profiles the households based on their association with different drivers of deforestation. Compared to the other household types, farming households had a larger family size and were affluent in terms of income and land property; however, they were slightly behind in terms of education status. Higher income and more land property among farming households were also observed in other deforestation-linked communities since the sale of agricultural products generates a comparatively high income ([Börner and Wunder, 2008](#); [Mullan et al., 2018](#)). Additionally, it is quite common for the family sizes of farming communities to be large. However, in this case, the lower education status of the farming community requires the attention of policymakers and relevant managers. Notably, only one-fourth of the betel leaf cultivation HHs had institutional education (76% with no formal education), which is significantly lower than the national average of Bangladesh.

While exploring the socioeconomic differences observed between the drivers and non-drivers, significant differences were found in terms of tenure, education, and annual income ([Table 2](#)). Comparatively, older houses with high settlement durations were found among the households related to deforestation drivers. Moreover, the houses of encroacher households were relatively new. Since the fuelwood collecting households and farming households had a much longer settlement duration, this resulted in the average tenure of drivers being significantly higher than that of non-drivers. Education was found to be lower among the drivers, which concurs with other studies noting that the households linked to direct deforestation have a low education status ([Mena et al., 2006](#); [Specht et al., 2019](#)). Having less education makes it difficult for households to engage in alternative income-generating activities; instead, they depend on forest resources. As a result, it is frequently seen that local communities with less education engage in

cultivation, fuelwood collection, and other forms of livelihood activities related to forest conversion. The households linked to deforestation had a lower annual income than those not directly linked to deforestation in TWS. This finding is in line with the findings of other studies suggesting that when a community is responsible for deforestation, it is relatively poor or has a lower average income (Angelsen and Wunder, 2003; Rivera et al., 2013; Specht et al., 2019; Tani, 2017b; Timko et al., 2010). After analyzing deforestation at the local level, it is evident that poverty within the community was one of the most important factors that must be addressed by forest managers and relevant policymakers. Communities that depend on forests and cause deforestation are typically poor, especially in developing countries. The relationship between poverty and deforestation must be understood by using appropriate context-specific empirical evidence.

5.2. Influence of socioeconomic context on deforestation

This study determined the influence of socioeconomic aspects on the drivers of deforestation by developing four logit models (Table 3). Among the studied variables, tenure, land, income, and occupation were statistically significant in at least three of the models. Tenure duration had an interesting influence on deforestation in the Teknaf forests. When considering the possibility of a household engaging in an activity that functions as a driver of deforestation, old households were more likely to do so than new ones. However, when only considering forest encroachment, households in new houses (settlement duration <5 years) were nearly 2.5 times more likely to do so than those in older houses. Similar findings were noted by Tani (2017), who found that new houses were encroaching on forests. In the case of the Teknaf forests, households with a higher tenure duration were engaging in betel leaf cultivation and fuelwood collection. On the contrary, the newer households were encroaching on forests. In other words, new households have become established within the forest reserve area, which is alarming for forest conservation initiatives and especially for PAs.

With regard to land property in Teknaf, small households were more engaged in deforestation than large households. The link between land property and deforestation is context-specific, and studies have found both positive and negative links between deforestation and land property (Mena et al., 2006; Yanai et al., 2020). In the case of TWS, the holding of a small land property can be linked to other drivers such as poverty and occupation. Small households are more likely to be poor and work as laborers for their livelihood. Both are linked to a greater likelihood of engaging in direct deforestation, resulting in a positive relationship between small land property and forest destruction.

In the case of Teknaf, poor households were more likely to engage in deforestation. Studies concerning deforestation in developing countries have often found a positive link between poverty and deforestation (Angelsen et al., 2014; Nerfa et al., 2020; Specht et al., 2019). Although poverty cannot be blamed as the single major cause of deforestation, this positive correlation should be considered by policymakers and forest managers. Additionally, a majority of the people living in or near forests in developing countries are estimated to live in extreme poverty (Shepherd, 2012). This is also seen in Teknaf, where approximately 60% of the population is poor. Poor people rely on forests more than others, and this dependency results in deforestation. In managing forests and developing conservation policies, these poor communities need to be included in forest management plans. If conservation plans do not include alternatives for an overdependency on forests, deforestation will continue as seen in the past.

With regard to occupation, farming and labor households are several times more likely to engage in deforestation. Notably, agriculture accounts for the majority of deforestation globally (Curtis et al., 2018; Geist and Lambin, 2001; Hosonuma et al., 2012). Several other regional studies have also confirmed a positive relationship between farming and deforestation (Alam et al., 2015; Tani and Rahman, 2017; Ullah et al., 2020a). In the case of Teknaf, farming and labor households were more

commonly engaged in the deforestation process than other households. Around TWS, people who own land perform farming, while those who do not have land become farm laborers. Betel leaves are one of the major cash crops cultivated in and around the PA, which is tied to the deforestation process (Ullah et al., 2020a). Besides betel leaf cultivation, people clear forest lands and cultivate rice in the valley of the hilly forest area. Farming and farmers are deeply rooted in the local community. While farming causes deforestation, its deep-rooted nature makes it a difficult activity to change. While taking initiatives to conserve forests, the local farming community should be considered one of the major driving forces of deforestation, while farming techniques that cause less harm to forests should be introduced.

The socioeconomic drivers influencing the deforestation process in the Teknaf Peninsula are interlinked. Due to poverty, some households have smaller land properties. Notably, those owning small lands cannot generate enough income to overcome poverty. As a result, these households have no alternative but to farm, collect fuelwood, or encroach on the forest to gain access to more land. Moreover, low education status and large family size cause the local community to become more strongly attached to the deforestation process. As such, the social structure of the Teknaf Peninsula needs to be addressed in forest management policies and regional development initiatives.

5.3. Role of forest management approaches: present and future

The BFD initiated a co-management approach in 2004 to manage the PAs of Bangladesh. This approach came into implementation through a pilot project that considered five PAs. Since then, the use of this approach has been growing in terms of the number of PAs and areas covered. Additionally, future plans related to the forest management conducted by the BFD indicate the continuation and scaling up of the co-management approach (BFD and CRPARP, 2016). However, studies have expressed concerns over the effectiveness of the current form of co-management implemented in Bangladesh. These concerns are in addition to those related to continued deforestation occurring within the PAs, weak governance, the ambiguity of specific rules and strategies, lack of participation from local communities, unequal benefit distribution, and low levels of transparency (Islam et al., 2020, 2018; Nishat and Chowdhury, 2019; Rahman and Islam, 2021). These concerns over the co-management approach are greater than the reported limited success in decreasing deforestation rates and improving the livelihoods of communities surrounding some PAs (Chowdhury et al., 2014; Islam et al., 2019). Notably, Mollick et al. (2021) highlighted a wide implementation gap between the ambitious co-management policy framework and the actual rollout executed on the ground by the BFD. In addition to the implementation gap of the co-management approach, the BFD has the capacity to enforce it. This study reported that, among the local community studied, 37% of households collected fuelwood, 11% encroached on the forest boundary, and 45% were involved in activities associated with at least one major driver of deforestation. The existence of these drivers of deforestation on a large scale indicates a limitation in the capacity of the BFD to enforce their policy. Studies have shown that even with clear intentions and initial implementation success, weak enforcement capacity can make the legitimacy of an entire forest conservation project questionable (Damnyag et al., 2012).

Moving beyond the limitations related to implementation and enforcement capacity in the co-management approach, the legal framework developed by the BFD has some significant weaknesses. The Protected Area Management Rule of 2017 was enacted on November 12, 2017. This rule has explicit legislative provisions and is the latest legal guideline for managing PAs under co-management (BFD, 2017). Moreover, it provides guidelines on organizing committees to manage PAs that include local communities. However, the concern is that there was no consideration of the power dynamics found among stakeholders, which is a prerequisite for active participation. Moreover, this rule does not have an adequate legal framework to ensure that local communities

have sufficient tenure rights. Additionally, no guidelines for easing community dependence on forests were provided. According to this study, 45% of the surveyed households were directly involved in the deforestation process; in other words, they were directly dependent on forests for their livelihoods. This has also been occurring in other PAs that are surrounded by local communities in Bangladesh. To implement effective forest co-management policies that benefit both forests and people, ensuring proper legal rights in relation to forests and the capacity development of local communities is necessary. Developing the capacity of local communities reduces forest dependence and ensures effective participation in the management process (Devkota, 2020; Rashid and Khan, 2014). Low education among the households strongly associated with deforestation in Teknaf reflects the importance of capacity development within the local community. Furthermore, proper tenure rights are an important principle for communities to manage natural resources (Ostrom, 1990). Secured property rights are widely understood as critical for the socioeconomic development and sustainable land management of forested areas (Miller et al., 2021). However, studies have found that co-management approaches cannot always ensure property rights, and there is evidence of reduced property rights following the implementation of the co-management approach (Hajjar et al., 2021). Therefore, forest policies related to the implementation of co-management in Bangladesh should specify and enforce the forest rights of local communities in such a way that they can simultaneously benefit from and protect forests. The PA (2017) rule gives local communities the right to collect non-timber forest products and hold 50% of the share of timber products (BFD, 2017). However, this type of right may instigate further deforestation and have little benefit for forest conservation.

6. Conclusion

This study identified, quantified, and qualified the relationships between the various socioeconomic determinants impacting deforestation in TWS. In the Teknaf region, poor people are heavily dependent on forests for fuelwood, agriculture, and land. This overdependency has translated into the proximate drivers of deforestation. In an attempt to explore the factors behind these drivers, this study found that poor people are more likely to engage in deforestation. In addition to households linked to deforestation drivers having a low annual income, they also have less education and land property. Moreover, farmers and laborers were more likely to engage in deforestation. Thus, households with a comparatively low socioeconomic status are causing higher levels of deforestation. Although this study was conducted in Teknaf, its results can be relevant to the management of other PAs in Bangladesh. If PAs are surrounded by communities with fragile social and economic structures, there is a high possibility of continued deforestation and the implementation of even stricter forest conservation policies.

The findings of this study reflect the situation in Bangladesh, where top-down forest conservation policies do not generate the expected outcomes. Even after the implementation of co-management for more than a decade, deforestation has occurred within PAs and their surrounding communities, resulting in no changes to their livelihoods or quality of life. Co-management aims to create an intricate relationship between local people and forests to provide sustainable outcomes for both. This type of relationship develops on the foundation of an appropriate legal framework and suitable management policies. The ability to model and project the course and effects of forest policies depends on the understanding of how deforestation occurs and how it might change in the future. This study sheds light on the complex interplay of socioeconomic factors influencing the deforestation process occurring within a specific PA in Bangladesh under existing forest policies. Based on the empirical findings of this study in relation to the impact of socioeconomic drivers on the deforestation process, we argue that there is significant potential to develop and implement policies and strategies that address these socioeconomic elements to conserve forests.

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Declaration of Competing Interest

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